

Name: Laiba Taseen
Reg No: 23-NTU-CS-1258

Project Title :

ESP32 Dual Switch - Dual LED using External Interrupts and Timer Debounce

Code:

```
/*  
  
Project Title : ESP32 Dual Switch - Dual LED using External Interrupts and Timer Debounce  
  
Author      : Laiba Taseen  
  
Reg No      : 23-NTU-CS-1258  
  
*/  
  
  
#include <Arduino.h>  
  
  
#define LED1_PIN 2  
#define LED2_PIN 4  
#define SW1_PIN 32  
#define SW2_PIN 14  
  
  
// Debounce time (milliseconds)  
#define DEBOUNCE_MS 50  
  
  
hw_timer_t *debounceTimer1 = NULL;  
hw_timer_t *debounceTimer2 = NULL;  
  
  
volatile bool sw1_pending = false;  
volatile bool sw2_pending = false;
```

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```
// Function declarations

void IRAM_ATTR sw1_isr();

void IRAM_ATTR sw2_isr();

void IRAM_ATTR debounceTimer1ISR();

void IRAM_ATTR debounceTimer2ISR();


void setup() {
    Serial.begin(115200);

    delay(100);

    Serial.println("ESP32 Dual Switch Dual LED Project - Core v3.x Compatible");


    pinMode(LED1_PIN, OUTPUT);
    pinMode(LED2_PIN, OUTPUT);
    digitalWrite(LED1_PIN, LOW);
    digitalWrite(LED2_PIN, LOW);


    pinMode(SW1_PIN, INPUT_PULLUP);
    pinMode(SW2_PIN, INPUT_PULLUP);


    debounceTimer1 = timerBegin(20000);
    debounceTimer2 = timerBegin(20000);


    // Attach ISRs

    timerAttachInterrupt(debounceTimer1, debounceTimer1ISR);
    timerAttachInterrupt(debounceTimer2, debounceTimer2ISR);
```

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```
attachInterrupt(digitalPinToInterrupt(SW1_PIN), sw1_isr, FALLING);  
attachInterrupt(digitalPinToInterrupt(SW2_PIN), sw2_isr, FALLING);  
}  
  
void loop() {  
  
    static unsigned long lastPrint = 0;  
    if (millis() - lastPrint > 2000) {  
        lastPrint = millis();  
        Serial.printf("LED1: %d | LED2: %d\n", digitalRead(LED1_PIN), digitalRead(LED2_PIN));  
    }  
}  
  
void IRAM_ATTR sw1_isr() {  
    detachInterrupt(digitalPinToInterrupt(SW1_PIN));  
    sw1_pending = true;  
  
    timerAlarm(debounceTimer1, DEBOUNCE_MS * 1000, false, 0);  
}  
  
void IRAM_ATTR sw2_isr() {  
    detachInterrupt(digitalPinToInterrupt(SW2_PIN));  
    sw2_pending = true;
```

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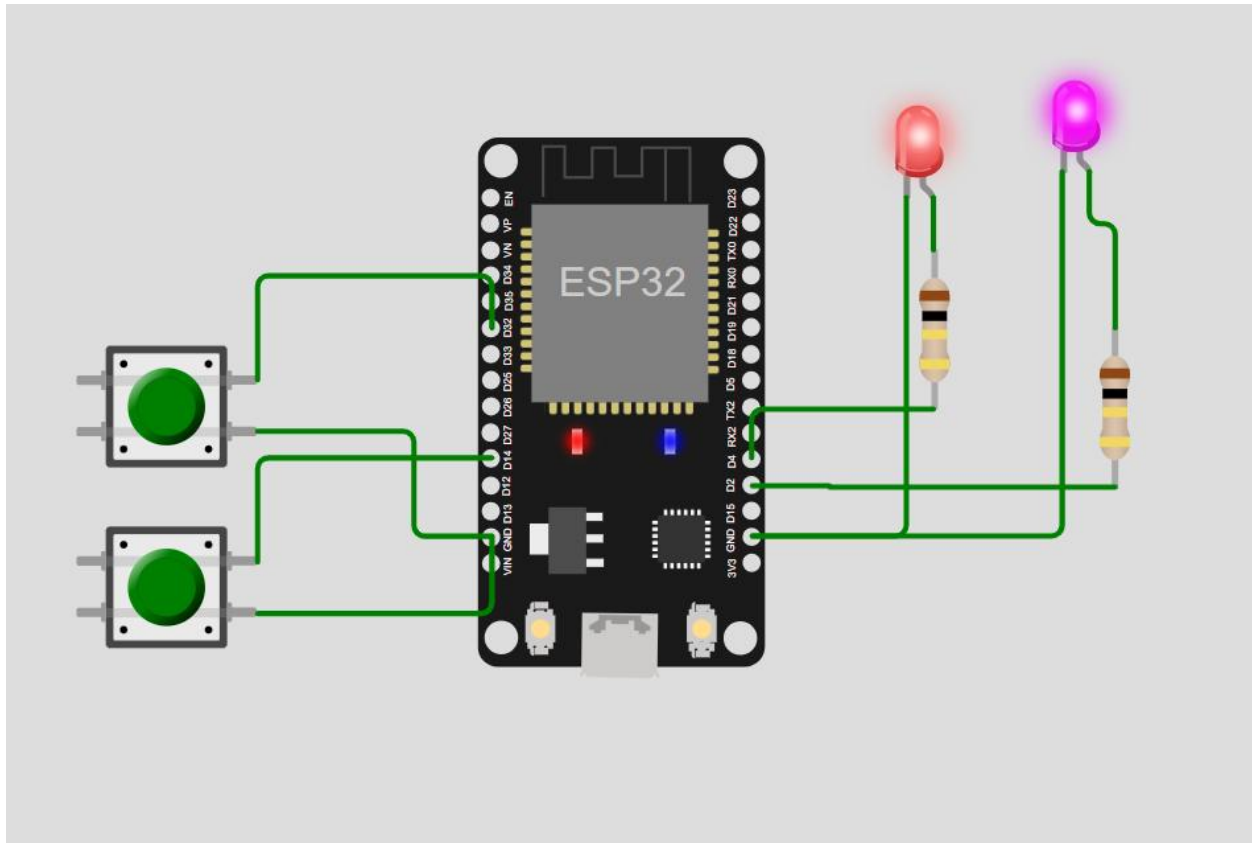
```
    timerAlarm(debounceTimer2, DEBOUNCE_MS * 1000, false, 0);
}

// Timer ISR for Switch 1
void IRAM_ATTR debounceTimer1ISR() {
    if (sw1_pending && digitalRead(SW1_PIN) == LOW) {
        digitalWrite(LED1_PIN, !digitalRead(LED1_PIN));
    }
    sw1_pending = false;
    attachInterrupt(digitalPinToInterrupt(SW1_PIN), sw1_isr, FALLING);
}

// Timer ISR for Switch 2
void IRAM_ATTR debounceTimer2ISR() {
    if (sw2_pending && digitalRead(SW2_PIN) == LOW) {
        digitalWrite(LED2_PIN, !digitalRead(LED2_PIN));
    }
    sw2_pending = false;
    attachInterrupt(digitalPinToInterrupt(SW2_PIN), sw2_isr, FALLING);
}
```

Output Screenshot:

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Wokwi Project Link:

<https://wokwi.com/projects/443984382660231169>